

Kevin Naveen

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EDUCATION

University of Ottawa

Honours Bachelor of Science, Specialization in Computer Science

Ottawa, ON

Sep 2021 – Apr 2026

LEADERSHIP

IEEE uOttawa Student Branch | McNaughton Centre Director

Sep 2025 – April 2026

- Planned and delivered **4 workshops** (AI, web, Git, embedded) and **10+** events for **200+ students** to promote technical and industry skills in the student body
- Managed **\$15,000** of shared lab hardware and organized lab sessions for student utilization

WORK EXPERIENCE

Emerdepot

Toronto, ON

Junior Software Engineer (Part-time)

June 2026 – Present

- Constructed a search service in **React** and **Express** for a **10,000+ item** Shopify catalog, replacing a paid external widget to improve storefront load reliability and reduce vendor lock-in.
- Implemented resilient SKU-matching logic to handle varied customer inputs (ignoring hyphens and letter case) and heavily weight exact matches, directly improving search accuracy for specific part numbers.
- Extending the search architecture with an **AI-powered** semantic layer, utilizing the **Shopify GraphQL API** and vector embeddings to surface relevant products beyond exact keyword matches.

Privy Council Office

Ottawa, ON

Software Engineering Intern

Sep 2025 – Dec 2025

- Primary developer on a full-stack delivery-tracking application, migrating it across major **Angular** and **.NET** versions and rebuilding the **ASP.NET Core API**, **SQL Server** data model, and **Angular** interface.
- Developed the **Angular/TypeScript** front end: reactive forms, sortable data tables, and filterable containers wired to server-side queries, packaged as reusable components reused across application screens.
- Implemented **ASP.NET Core** services and endpoints for the application's core workflows, then migrated pagination, filtering, and sorting from the client into **SQL Server** queries, cutting render time on high-volume views by **60%**.
- Introduced **Angular signals** to the team and refactored shared components from **RxJS** subscription handling to signal-based state, removing manual lifecycle teardown and standardizing new feature work.

Software Analyst Intern

Sep 2024 – Dec 2024

- Automated** employee onboarding by building an **SSIS ETL** package that moved **500+** employee records from the personnel database into the department's user registry, cutting manual effort by an estimated **70%** per cycle.
- Extended **SSIS** packages with custom **C#** scripting for workflow-specific validation, data transformation, and automation.
- Built a second **ETL** pipeline in **Python** and **Pandas** that cleans, deduplicates, and normalizes Excel datasets submitted across the department, then batch-loads the structured records into **SQL Server**.

PROJECTS

AI/ML Autonomous Drone Hover and Recovery System | Python, MuJoCo, Gymnasium, PyTorch, Stable-Baselines3

- Led a team of 3** from simulation design to hardware deployment, designing the training pipeline and **flight-test protocol**, environment, evaluation, and firmware workstreams across the team.
- Developed a **PPO** sim-to-real reinforcement learning policy for autonomous hover control of a **Bitcraze Crazyflie 2.1** nano-quadrotor in **MuJoCo**, using custom reward shaping and early termination to hold a 1 m target within a ± 10 cm band.
- Engineered **domain randomization**, observation normalization, and **16-environment** parallel training infrastructure to improve policy robustness and generalization across **40M timesteps**, raising sim-to-real transfer to a **75% success rate**.
- Deployed the trained policy to physical hardware through a **cflib** interface, calibrating simulated thrust to firmware counts and validating stable hover and recovery from 25° tilt perturbations across live flight tests.

Real-Time Messaging System | Node.js, Redis, WebSockets, PostgreSQL, React

- Architected real-time chat infrastructure on native **WebSockets** with **Redis Pub/Sub** for cross-server broadcasting, using an in-memory room-to-connection registry.
- Built the REST API from scratch on **node:http** and native **HTTPS/TLS** with **no external frameworks**, writing a regex-compiled **dynamic routing engine** with named capture groups and custom validation middleware.
- Hardened API security with stateless **JWT** authentication, hashed refresh-token rotation, and signed **cursor-based pagination** that rejects client-tampered cursors.
- Building the **React** front end: a WebSocket client for live message sync and infinite scroll over the paginated API.

SKILLS

Languages: C#, TypeScript, JavaScript, Python, Java, C/C++, SQL, HTML/CSS

Frameworks: Angular, React, ASP.NET Core, Node.js, Express, PyTorch, Stable-Baselines3, Gymnasium, MuJoCo

Data & Tools: PostgreSQL, SQL Server, Redis, SSIS, Pandas, Scikit-Learn, Docker, Git, Azure DevOps